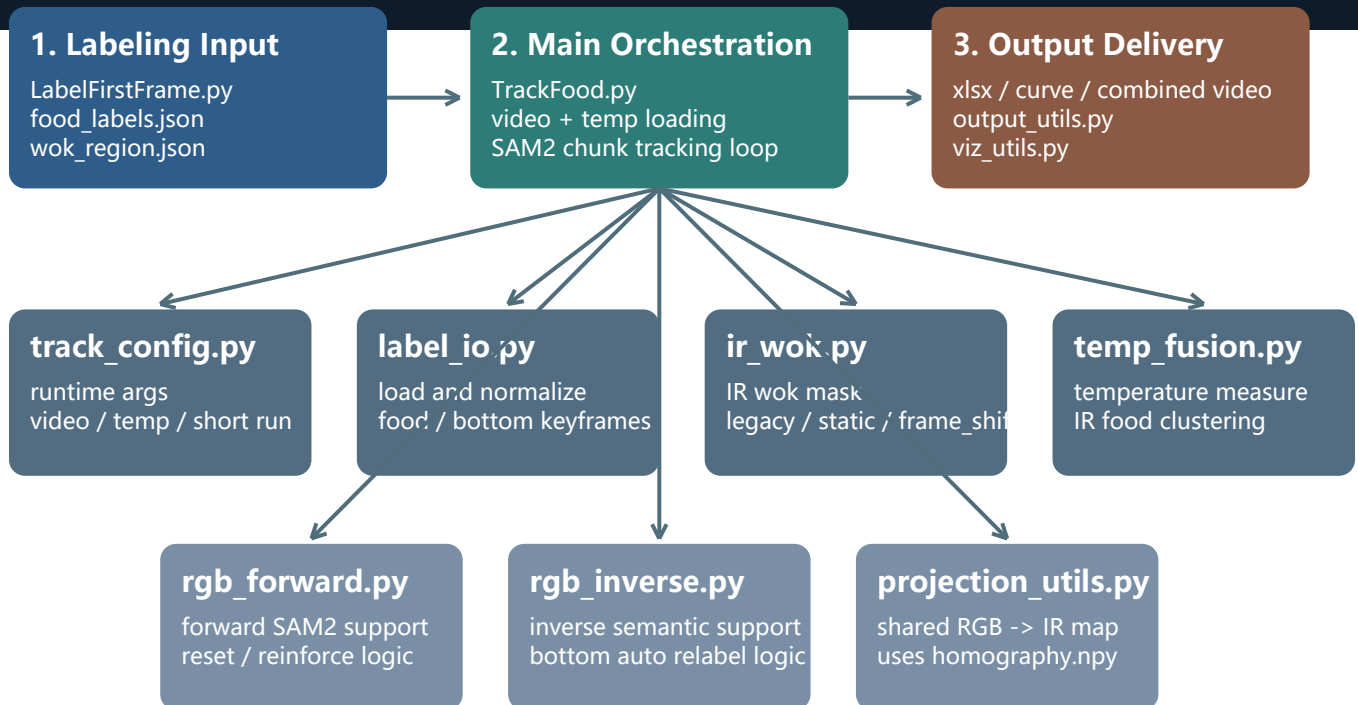


Chef Vision Current Architecture

Current decoupled workflow after modular refactoring

Focus: RGB forward / RGB inverse / IR thermal clustering collaboration



Module Summary And Current Position

Core workflow

- Manual labeling initializes RGB food, RGB bottom, and IR wok region.
- TrackFood.py now acts as the orchestrator instead of holding every utility.
- RGB forward, RGB inverse, and IR thermal signals work as complementary paths.

Current strengths

- Multi-modal constraints improve usability over a single RGB-only pipeline.
- Temperature curves, Excel logs, and combined videos are produced in one run.
- IR wok region logic is isolated, so future strategy changes are safer to test.

Current limitation

- Tracking precision is usable but not yet fully stable in highly dynamic scenes.
- The next high-value work should focus on algorithm stability rather than more refactoring.

Key module roles

- track_config.py: runtime path and parameter entry.
- label_io.py: standardized label loading.
- ir_wok.py: IR wok region strategy hub.
- rgb_forward.py / rgb_inverse.py: two RGB-side support strategies.
- temp_fusion.py: all temperature-related measurements.
- projection_utils.py: shared RGB-to-IR projection based on homography.
- output_utils.py + viz_utils.py: presentation layer and deliverables.

Main program status

- TrackFood.py is much cleaner than before, but it is still the core scheduler.
- Infrastructure cleanup is mostly done; future gains now come from algorithm tuning.